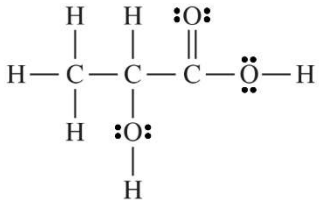
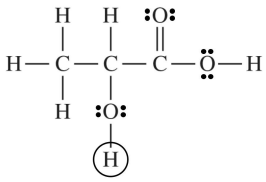
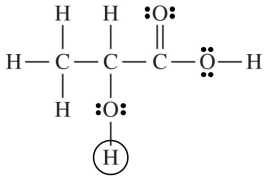
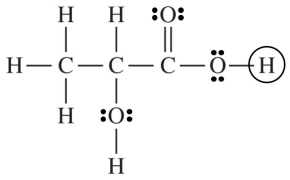
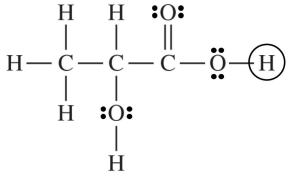
**Topics 8.6 – 8.11: MCQ Practice**



Carboxylic acids are one common class of weak acid. The structural formula of lactic acid, $HC_3H_5O_3$, is shown above and is an example of a carboxylic acid. Lactic acid undergoes an acid-base reaction with hydroxide ions according to the equation shown below.

$$HC_3H_5O_3(aq) + OH^-(aq) \rightarrow C_3H_5O_3^-(aq) + H_2O(l)$$

Which of the following correctly identifies the hydrogen atom in the lactic acid molecule that most readily participates in the reaction with the hydroxide ion and provides the correct justification?

| | Hydrogen Atom that Most Readily Participates in the Reaction with $OH^-(aq)$ | Justification |
|---|---|---|
| (A) |  | The $C_3H_5O_2^-$ ion is made more stable by the localization of the negative charge on a single oxygen atom. |
| (B) |  | The $C_3H_5O_2^-$ ion is made more stable by the delocalization of the negative charge on two different oxygen atoms through resonance. |
| (C) |  | The $C_3H_5O_2^-$ ion is made more stable by the localization of the negative charge on a single oxygen atom. |
| (D) |  | The $C_3H_5O_2^-$ ion is made more stable by the delocalization of the negative charge on two different oxygen atoms through resonance. |



2. The Lewis electron-dot diagrams of the HOCl molecule and the HOI molecule are shown above. Which of the following identifies the substance that represents the stronger acid and correctly identifies a factor that contributes to its being the stronger acid?

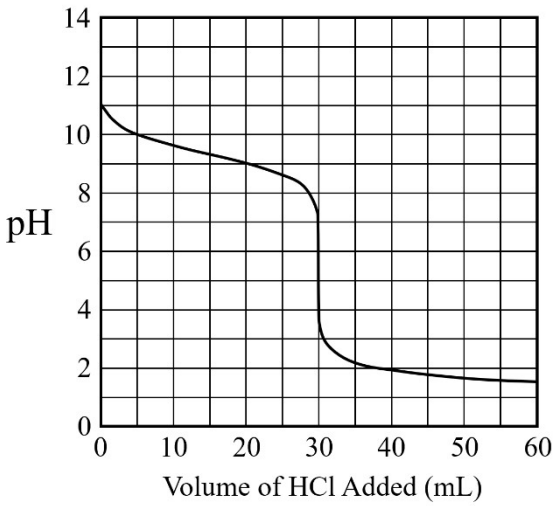
| | Stronger acid | Factor that contributes to its being the stronger acid |
|---|---|---|
| (A) | HOCl(*aq*) | HOCl molecules experience weaker London dispersion forces. |
| (B) | HOCl(*aq*) | The chlorine atom is more electronegative than the iodine atom, and this helps to stabilize the conjugate base. |
| (C) | HOI(*aq*) | HOI molecules experience stronger London dispersion forces. |
| (D) | HOI(*aq*) | The iodine atom is more electronegative than the chlorine atom, and this helps to stabilize the conjugate base. |

| Properties of Bromothymol Blue | | |
|---|---|---|
| low pH color | pH range of color change | high pH color |
| yellow | 6.0 – 7.6 | blue |

3. Information about the properties of the indicator bromothymol blue is listed in the table above. A student uses bromothymol blue to test a solution in a beaker containing a mixture of HOCl(*aq*) ($pK_a = 7.54$) and NaOCl(*aq*). The color of the solution turns yellow after a few drops of bromothymol blue are added to the beaker. Based on the results of this experiment, which of the following is true about the pH of the solution in the beaker and the relative concentrations of HOCl(*aq*) and $OCl^-(aq)$?

| | pH of the Solution | Relative Concentrations of HOCl(*aq*) and $OCl^-(aq)$ |
|---|---|---|
| (A) | Less than 6.0 | $[HOCl] > [OCl^-]$ |
| (B) | Less than 6.0 | $[HOCl] < [OCl^-]$ |
| (C) | Greater than 7.6 | $[HOCl] > [OCl^-]$ |
| (D) | Greater than 7.6 | $[HOCl] < [OCl^-]$ |

A sample of $NH_3(aq)$ is titrated with HCl($aq$). The pH of the solution is measured with a pH meter and graphed as a function of the volume of HCl($aq$) added, as shown in the graph at right.




A student carries out the same titration, but wants to use an indicator instead of a pH meter. Information about two different indicators is listed in the table below.

| Indicator | p$K_a$ |
|---|---|
| Methyl red | 5.0 |
| Phenolphthalein | 9.4 |

4. Which of the following identifies the best indicator to use for the titration and provides the correct justification?

| | Indicator | Justification |
|---|---|---|
| (A) | Methyl red | The p$K_a$ value of methyl red is close to the pH of the solution at the half-equivalence point. |
| (B) | Methyl red | The p$K_a$ value of methyl red is close to the pH of the solution at the equivalence point. |
| (C) | Phenolphthalein | The p$K_a$ value of phenolphthalein is close to the pH of the solution at the half-equivalence point. |
| (D) | Phenolphthalein | The p$K_a$ value of phenolphthalein is close to the pH of the solution at the equivalence point. |

5. A solution is prepared by combining equal volumes of 1.0 *M* HF($aq$) and 1.0 *M* NaF in a beaker. Which of the following correctly predicts how the pH will change if a small amount of HCl($aq$) or NaOH($aq$) is added to the beaker and provides the correct equation for the reaction that accounts for the relatively small change in pH?

| | Substance Added | How Will the pH Change? | Equation for the Reaction that Accounts for the Relatively Small Change in pH |
|---|---|---|---|
| (A) | HCl($aq$) | Decrease Slightly | $H_3O^+(aq) + F^-(aq) \rightarrow HF(aq) + H_2O(l)$ |
| (B) | HCl($aq$) | Increase Slightly | $Cl^-(aq) + HF(aq) \rightarrow HCl(aq) + F^-(aq)$ |
| (C) | NaOH($aq$) | Decrease Slightly | $OH^-(aq) + HF(aq) \rightarrow F^-(aq) + H_2O(l)$ |
| (D) | NaOH($aq$) | Increase Slightly | $OH^-(aq) + F^-(aq) \rightarrow HF(aq) + O^{2-}(aq)$ |

$$C_2H_5NH_2(aq) + H_2O(l) \rightleftarrows C_2H_5NH_3^+(aq) + OH^-(aq)$$

6. Ethylamine, $C_2H_5NH_2$, reacts with water according to the equation shown above. A buffer solution is prepared that contains equimolar amounts of $C_2H_5NH_2(aq)$ and $C_2H_5NH_3^+(aq)$. When a small amount of 12 *M* $HNO_3(aq)$ is added to this buffer solution, the pH of the solution changes from 10.75 to 10.73. Which of the following equations represents the reaction that accounts for the fact that the pH does not change significantly when $HNO_3(aq)$ is added?

   (A) $C_2H_5NH_2(aq) + H^+(aq) \rightarrow C_2H_5NH_3^+(aq)$

   (B) $C_2H_5NH_3^+(aq) + H^+(aq) \rightarrow C_2H_5NH_4^{2+}(aq)$

   (C) $NO_3^-(aq) + H^+(aq) \rightarrow HNO_3(aq)$

   (D) $OH^-(aq) + H^+(aq) \rightarrow H_2O(l)$

$$HCOOH(aq) + H_2O(l) \rightleftarrows H_3O^+(aq) + HCOO^-(aq)$$

7. Methanoic acid, HCOOH, reacts with water according to the equation shown above. A buffer solution is prepared that contains equimolar amounts of $HCOOH(aq)$ and $HCOO^-(aq)$. When a small amount of 12 *M* $KOH(aq)$ is added to this buffer solution, the pH of the solution changes from 3.74 to 3.76. Which of the following equations represents the reaction that accounts for the fact that the pH does not change significantly when $KOH(aq)$ is added?

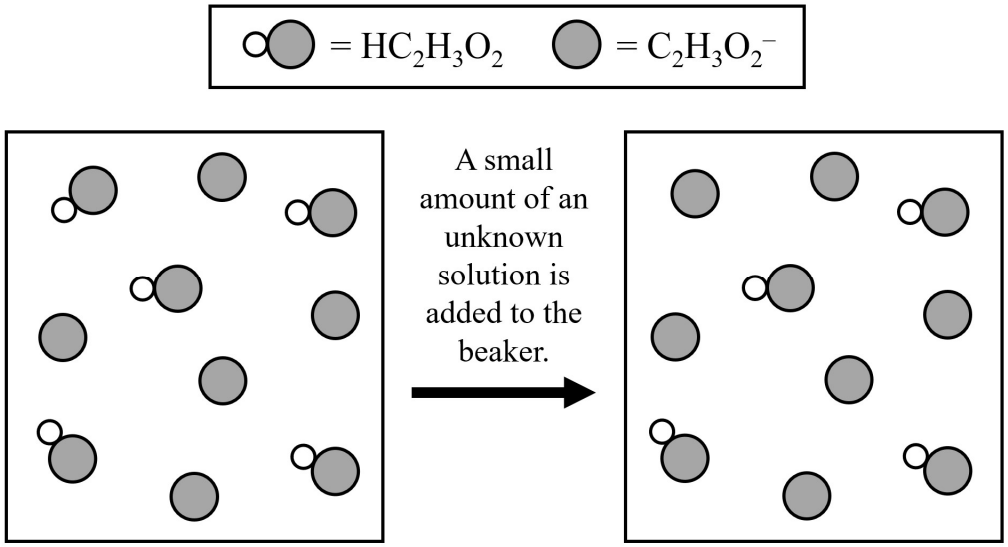
   (A) $HCOO^-(aq) + OH^-(aq) \rightarrow HCOOH(aq) + O^{2-}(aq)$

   (B) $HCOOH(aq) + OH^-(aq) \rightarrow HCOO^-(aq) + H_2O(l)$

   (C) $K^+(aq) + OH^-(aq) \rightarrow KOH(aq)$

   (D) $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$

8. A student measures the pH of a solution in a beaker that contains equimolar amounts of $HC_2H_3O_2(aq)$ and $NaC_2H_3O_2(aq)$ and records the pH as 4.74. Then the student adds a small amount of an unknown solution to the beaker and stirs the mixture thoroughly. The identity of the unknown solution is either HCl(*aq*) or NaOH(*aq*). The particulate diagrams shown below represent the relative amounts of $HC_2H_3O_2$ and $C_2H_3O_2^-$ in a sample of the solution in the beaker before and after the unknown solution is added.
(Note that water molecules and ions such as $Na^+$ or $Cl^-$ are omitted.)




Which of the following correctly identifies the unknown solution that is added to the beaker and describes the pH of the mixture at the end of the experiment?

| | Identity of the Unknown Solution | pH of the Mixture at the End of the Experiment |
|---|---|---|
| (A) | HCl(*aq*) | Less than 4.74 |
| (B) | HCl(*aq*) | Greater than 4.74 |
| (C) | NaOH(*aq*) | Less than 4.74 |
| (D) | NaOH(*aq*) | Greater than 4.74 |

| | $K_a$ |
|---|---|
| $HC_3H_5O_3(aq)$ | $8.3 \times 10^{-4}$ |
| $CH_3NH_3^+(aq)$ | $2.3 \times 10^{-11}$ |

9. The acid dissociation constants of $HC_3H_5O_3(aq)$ and $CH_3NH_3^+(aq)$ are given in the table above. Which of the following mixtures would produce a buffer solution with a pH of approximately 3 ?

   (A) A mixture of 100. mL of 0.10 *M* $HC_3H_5O_3$ and 50. mL of 0.10 *M* NaOH

   (B) A mixture of 100. mL of 0.10 *M* $CH_3NH_3Cl$ and 50. mL of 0.10 *M* NaOH

   (C) A mixture of 100. mL of 0.10 *M* $HC_3H_5O_3$ and 100. mL of 0.10 *M* NaOH

   (D) A mixture of 100. mL of 0.10 *M* $CH_3NH_3Cl$ and 100. mL of 0.10 *M* NaOH

10. A student attempts to prepare a buffer solution with a pH of 7.5 by adding HOCl ($pK_a = 7.54$) and NaOCl to distilled water. Instead of the desired pH, the measured pH of the buffer solution is 7.0. Which of the following is the most likely reason that the pH is too low?

    (A) Too much HOCl was used.

    (B) Too much NaOCl was used.

    (C) There was water in the flask before HOCl and NaOCl were added.

    (D) In order to prepare the buffer, a few drops of a strong base should have been added.

11. The $K_a$ value of hydrofluoric acid, HF(*aq*), is equal to $6.8 \times 10^{-4}$. Which of the following mixtures would produce a solution with a pH that is closest to 3.5?

    (A) 50.0 mL of 1.0 *M* HF(*aq*) and 50.0 mL of 1.0 *M* NaF(*aq*)

    (B) 50.0 mL of 1.0 *M* HF(*aq*) and 50.0 mL of 2.0 *M* NaF(*aq*)

    (C) 50.0 mL of 1.0 *M* HF(*aq*) and 50.0 mL of 1.0 *M* NaOH(*aq*)

    (D) 50.0 mL of 2.0 *M* HF(*aq*) and 50.0 mL of 1.0 *M* NaF(*aq*)

12. The $K_a$ value of acetic acid, $HC_2H_3O_2$, is equal to $1.8 \times 10^{-5}$. A buffer solution is prepared by mixing 500. mL of 2.0 *M* $HC_2H_3O_2(aq)$ and 500. mL of 1.0 *M* $NaC_2H_3O_2(aq)$. The pH of this buffer solution is closest to

(A) 4.44

(B) 4.74

(C) 5.05

(D) 7.00

| Solution | $[HNO_2]$ | $[NO_2^-]$ |
|---|---|---|
| #1 | 0.10 *M* | 0.10 *M* |
| #2 | 1.0 *M* | 1.0 *M* |

13. Two buffer solutions are prepared, and the concentrations of each buffer component are listed in the table above. Which of the following correctly compares the pH and the buffer capacity of solutions #1 and #2?

| | Comparison of pH | Comparison of Buffer Capacity |
|---|---|---|
| (A) | Solutions #1 and #2 have the same pH. | Solutions #1 and #2 have the same buffer capacity. |
| (B) | Solutions #1 and #2 have the same pH. | Solution #2 has a larger buffer capacity than solution #1. |
| (C) | Solution #2 has a higher pH than solution #1. | Solutions #1 and #2 have the same buffer capacity. |
| (D) | Solution #2 has a higher pH than solution #1. | Solution #2 has a larger buffer capacity than solution #1. |

| Solution | Buffer Components | Initial pH | Amount of 10.0 *M* $HNO_3(aq)$ Added to 1.00 L of the Buffer Solution | Final pH After the Addition of $HNO_3(aq)$ |
|---|---|---|---|---|
| X | $NH_3(aq)$ and $NH_4^+(aq)$ | 9.3 | 10.0 mL | 9.2 |
| Y | $NH_3(aq)$ and $NH_4^+(aq)$ | 9.0 | 10.0 mL | 2.2 |

14. Two different buffer solutions X and Y are prepared by combining solutions of $NH_3(aq)$ and $NH_4^+(aq)$. A student records the initial pH of each buffer solution and then adds 10.0 mL of 10.0 *M* $HNO_3(aq)$ to 1.00 L of each buffer solution. The student correctly measures the final pH of each solution. Data from the experiment are shown in the table above. Which of the following correctly compares the buffer capacity of solutions X and Y and provides the correct justification?

| | Comparison of Buffer Capacity | Justification |
|---|---|---|
| (A) | Solution X has a larger buffer capacity than solution Y. | The initial pH of solution X is slightly higher than the initial pH of Solution Y. |
| (B) | Solution X has a larger buffer capacity than solution Y. | The change in pH for solution X is much smaller than the change in pH for solution Y. |
| (C) | Solution Y has a larger buffer capacity than solution X. | The initial pH of solution Y is slightly lower than the initial pH of Solution X. |
| (D) | Solution Y has a larger buffer capacity than solution X. | The change in pH for solution Y is much larger than the change in pH for solution X. |

$$Mg(OH)_2(s) \rightleftarrows Mg^{2+}(aq) + 2\ OH^-(aq)$$

15. The dissolution of $Mg(OH)_2(s)$ in water is represented by the equation above. The $K_{sp}$ of $Mg(OH)_2$ is $5.6 \times 10^{-12}$. Which of the following changes will increase the solubility of $Mg(OH)_2$ in an aqueous solution?

(A) Decreasing the pH

(B) Increasing the pH

(C) Adding $NH_3$ to the solution

(D) Adding $Mg(NO_3)_2$ to the solution

$$CaF_2(s) \rightleftarrows Ca^{2+}(aq) + 2\ F^-(aq)$$

16. The dissolution of $CaF_2(s)$ in water is represented by the equation above. The $K_{sp}$ of $CaF_2$ is $3.9 \times 10^{-11}$. The molar solubility of $CaF_2$ increases when it is dissolved in 0.50 $M$ $HNO_3(aq)$ instead of neutral distilled water. Which of the following represents the balanced net-ionic equation for the process that occurs between species in solution that contributes to the increased solubility of $CaF_2(aq)$ in $HNO_3(aq)$?

(A) $F^-(aq) + H_3O^+(aq) \rightleftarrows HF(aq) + H_2O(l)$

(B) $2\ F^-(aq) + 2\ NO_3^-(aq) \rightleftarrows F_2(g) + N_2(g) + 3\ O_2(g)$

(C) $Ca^{2+}(aq) + 2\ F^-(aq) \rightleftarrows CaF_2(s)$

(D) $Ca^{2+}(aq) + 2\ NO_3^-(aq) \rightleftarrows Ca(NO_3)_2(s)$